Edwin Kayang

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www.pelpuo.github.io/ekayang-portfolio/

Education

Arizona State University *August 2023 – Present*
PhD Student in Computer Science

University of Ghana *Aug 2018 – Mar 2023*
BS in Computer Engineering

- CGPA: 3.75/4.0

Technologies

Languages: C, C++, Java, RISC-V Assembly, Python, JavaScript, Rust

Technologies: LLVM, Intel Pin, Tensorflow, PyTorch, Git, Cmake, GDB, Qemu, React/ReactNative, NodeJS, FastAPI, Flask

Research Interests

My research interests broadly lie in the area of program analysis and transformation. Through compilers, binary analysis and instrumentation, and profile-guided optimization, my research seeks to develop techniques that better integrate program behavior and hardware characteristics into the compilation pipeline, enabling software to adapt to the requirements, constraints, and capabilities of emerging applications and computing systems.

Professional Experience

Research Assistant *Tempe, AZ*
Secure, Trusted and Assured Microelectronics (STAM) Center, Arizona State University *Aug 2023 – Present*

- Developing program analysis and compilation technologies aimed at ensuring memory safety in legacy applications, while maintaining performance.
- Investigating algorithms to facilitate profile-guided optimization and compiler autotuning.

Innovations Graduate Executive *Accra, Ghana*
4th-IR *June 2022 – Aug 2023*

- Developed and integrated application programming interfaces (APIs) for machine learning models into a model benchmarking platform.
- Maintained local GPU servers for training, testing, and deploying machine learning models.
- Conducted a comprehensive survey of machine learning models, providing rankings and recommendations for 20 distinct tasks.

SuaCode Fellow *Remote*
Nsesa Foundation *June 2021 – Sept 2021*

- Developed, tested, and evaluated components of an automated grading tool for online courses, supporting scalable assessment and feedback
- Authored technical documentation describing the system design, implementation, and experimental evaluation

Teaching Experience

Teaching Assistant, Arizona State University

CSE 355: Introduction to Theoretical Computer Science	Fall 2026
CSE 478: Foundations of Data Visualization	Spring 2026

Publications

EnCert: Compiler-Based Formal Verification of Namespace-Protected Applications (Under Review)
Edwin Kayang, Mishel Jyothis Paul, Michel A. Kinsy
International Conference on Verification, Model Checking, and Abstract Interpretation

MOSS: Mixed-Policy Optimization for Speed Under Hard Binary-Size Constraints (Under Review)
Edwin Kayang, Mishel Jyothis Paul, Nges Brian Njungle, Michel A. Kinsy
Design, Automation and Test in Europe Conference 2027

EnForge: Enforcing Memory Safety in Legacy Applications through Capability Injection (Just Accepted)
Edwin Kayang, Mishel Jyothis Paul, Michel A. Kinsy
Asian Hardware Oriented Security and Trust Symposium

ARCHE: Hardware-Managed Pointer Provenance Propagation for Capability Architectures (Just Accepted)
Mishel Jyothis Paul, ***Edwin Kayang***, Eric Jahns, Michel A. Kinsy
Asian Hardware Oriented Security and Trust Symposium

FLAVIUM: A Multi-Agent System for Reactive Defensive Cyber Operations with Distilled Neural Policies (Just Accepted)
Luigi Mastromauro, Yaphet Elias Weldegebriel, Mishel Jyothis Paul, ***Edwin Kayang***, Muslum Ozgur Ozmen, Michel A. Kinsy
IEEE High Performance Extreme Computing

DALC-CT: Dynamic Analysis of Low-Level Code Traces for Constant-Time Verification Oct 2025
Nges Brian Njungle, ***Edwin Kayang***, Mishel Jyothis Paul, Michel A. Kinsy
arXiv preprint arXiv:2604.16832

Panoptes: A Framework for Profile Clustering and Context-Aware Binary Optimization Dec 2025
Edwin Kayang, Eric Jahns, Mishel Jyothis Paul, Michel A. Kinsy
ACM/IEEE International Conference on Software Engineering
doi.org/10.1145/3744916.3787767

Prismo: A Decision Support System for Privacy-Preserving ML Framework Selection Oct 2025
Nges Brian Njungle, Eric Jahns, Luigi Mastromauro,***Edwin Kayang***, Milan Stojkov, Michel A. Kinsy
arXiv preprint arXiv:2510.09985

AQUILA: A Flexible Architecture Guideline for Building Custom Distributed Systems Testbeds *(Just Accepted)* Sept 2025
Luigi Mastromauro, ***Edwin Kayang***, Mishel Jyothis Paul, Eric Jahns, Muslum Ozgur Ozmen, Michel A. Kinsy
IEEE/IFIP International Conference on Embedded and Ubiquitous Computing
doi.org/10.1109/EUC66494.2025.00009

R-Visor: An Extensible Dynamic Binary Instrumentation and Analysis Framework for Open Instruction Set Architectures June 2025
Edwin Kayang, Mishel Jyothis Paul, Eric Jahns, Muslum Ozgur Ozmen, Milan Stojkov, Kevin Rudd, Michel A. Kinsy
ACM SIGPLAN/SIGBED Conference on Languages Compilers Tools and Theory for Embedded Systems
doi.org/10.1145/3735452.3735522

Autograd: Automated grading software for mobile game assignments in Suacode courses Nov 2021
Prince Steven Annor, ***Edwin Kayang***, Samuel Boateng, George Boateng
Computer Science Education Research Conference
doi.org/10.1145/3507923.3507954

Research Artifacts

R-Visor: An extensible Dynamic Binary Instrumentation (DBI) and analysis framework with initial support for the RISC-V Instruction Set Architecture (ISA). Oct 2025
Documentation

Panoptes: A region-based program similarity analysis framework. Mar 2026
Tool

Invited Talks

RAIL: RISC-V Analysis and Instrumentation Library Apr 2024
Latch-Up 2024, The FOSSi Foundation

Professional Affiliations

Tribe Lead, African PhD Students Network at ASU Aug 2025 - Aug 2026
- Organized social events to connect PhD students and alumni of Arizona State University

Shadow PC Reviewer, International Conference on Software Engineering (ICSE) Aug 2025 - Oct 2025
- Gave comprehensive reviews for 7 papers submitted to ICSE 2026

Machine Learning, DevOps and 3D Modeling Instructor, Trestle Academy, Ghana June 2023 - Aug 2023
- Taught DevOps technologies, Machine learning with Tensorflow, and 3D modeling with Blender to roughly 30 students.

IT/Publications Head, Enactus, University of Ghana Oct 2021 - June 2023
- Led a team of web and mobile app developers to create solutions for the Sustainable Development Goals

Lead, Google Developer Student Clubs, University of Ghana Oct 2020 - Aug 2021
- Organized tech events focused on teaching skills and exposing upcoming software developers to industry practices.